ME340 Elasticity and Inelasticity – Wei Cai – Stanford University – Win 2026

Handout 14. Graphical Representation

February 9, 2026

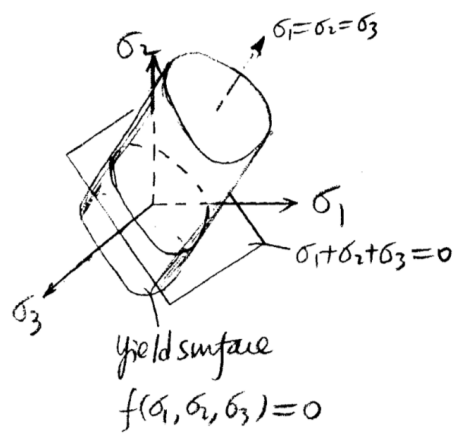
In this lecture, we give more discussions on the yield conditions (Von Mises and Tresca), including their graphical representation and experimental verification.

1 Yield surface in the space of principal stresses

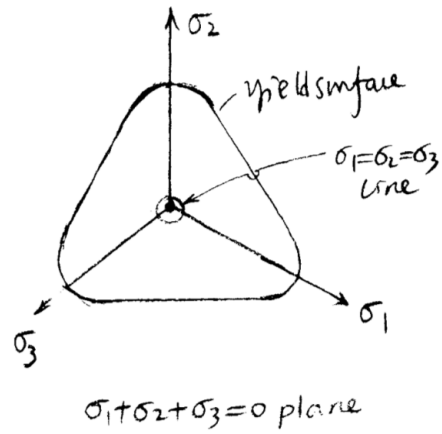
If the material is isotropic, the three principal stress values contain all the information about whether the material has reached the yield condition or not. In other words, the yield condition is reached on a 2D surface, i.e., the yield surface, in the 3D space spanned by $\sigma_1, \sigma_2, \sigma_3$:

$$f(\sigma_1, \sigma_2, \sigma_3) = 0 \tag{1}$$

Furthermore, experiments have shown that hydrostatic stress $\bar{\sigma} = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$ plays no role in yielding. This means that the yield surface $f(\sigma_1, \sigma_2, \sigma_3) = 0$ must have a prismatic shape, with the axis defined by the line $\sigma_1 = \sigma_2 = \sigma_3$ (i.e., the [1 1 1] direction).



In other words, the intersection between the yield surface and any plane defined by $\sigma_1 + \sigma_2 + \sigma_3 = c$ gives a curve of the same shape. Imagine looking down the $\sigma_1 = \sigma_2 = \sigma_3$ line; the entire yield surface looks like a curve.



This means that to construct the yield surface, we only need to specify the shape of this curve and translate it along the [1 1 1] axis. Material isotropy requires the yield surface to be symmetric with respect to 120° rotation around the $\sigma_1 = \sigma_2 = \sigma_3$ line (i.e., $\sigma_1 \to \sigma_2, \sigma_2 \to \sigma_3, \sigma_3 \to \sigma_1$) as well as mirror reflection (e.g., $\sigma_1 \leftrightarrow \sigma_3$).

Other geometric restrictions can be imposed on the yield surface. For example, the yield surface is usually assumed to be convex, which we will discuss later.

2 Von Mises yield condition

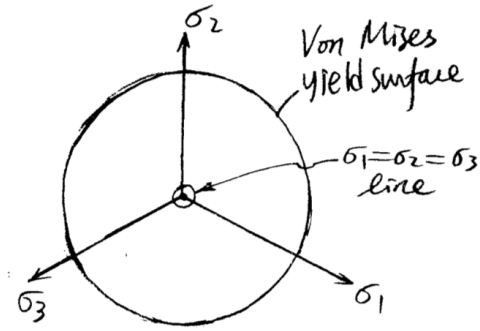
The Von Mises yield condition is defined as:

$$J_2 \equiv \frac{1}{2}\left(s_1^2 + s_2^2 + s_3^2\right) = k^2 \tag{2}$$

where s_1, s_2, s_3 are the principal stresses of the deviatoric stress s_{ij}. Recall that $\bar{\sigma} = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$, and the principal values of the deviatoric stress are $s_1 = \sigma_1 - \bar{\sigma}$, $s_2 = \sigma_2 - \bar{\sigma}$, $s_3 = \sigma_3 - \bar{\sigma}$. Substituting these expressions into Eq. (2), the Von Mises yield condition can be written as,

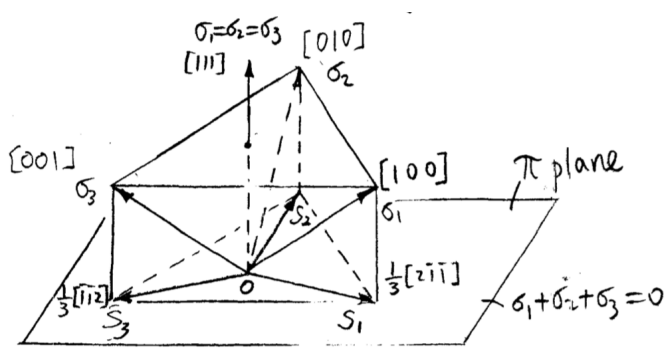
$$\begin{aligned}
k^2 = J_2 &= \frac{1}{2}\left[(\sigma_1 - \bar{\sigma})^2 + (\sigma_2 - \bar{\sigma})^2 + (\sigma_3 - \bar{\sigma})^2\right] \\
&= \frac{1}{2}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 3\,\bar{\sigma}^2\right] \\
&= \frac{1}{3}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - (\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)\right] \\
&= \frac{1}{6}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]
\end{aligned}$$

The Von Mises yield surface is a cylinder with a circular cross-section. The proof is given in the following.



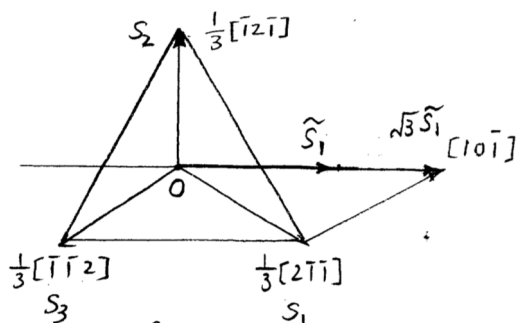
Geometric Proof

To be precise, we express the condition in terms of vectors on the π-plane ($\sigma_1+\sigma_2+\sigma_3=0$). Note that $s_1+s_2+s_3=0$; so this is also the plane of s_1, s_2, s_3.



The principal stresses s_1, s_2, s_3 can be obtained from σ_1, σ_2, σ_3 as follows.

$$s_1=\frac{2\sigma_1-\sigma_2-\sigma_3}{3},\quad s_2=\frac{-\sigma_1+2\sigma_2-\sigma_3}{3},\quad s_3=\frac{-\sigma_1-\sigma_2+2\sigma_3}{3} \tag{3}$$



Since s_1, s_2, s_3 are not independent, we introduce $\tilde{s}_1$ perpendicular to the axis of s_2 on the π-plane:

$$\tilde{s}_1=\frac{\sigma_1-\sigma_3}{\sqrt{3}}=\frac{s_1-s_3}{\sqrt{3}} \tag{4}$$

Our task is to express the Von Mises yield criterion in terms of $\tilde{s}_1$ and s_2 and show that it is a circle. In terms of $\tilde{s}_1$ and s_2,

$$s_1=\frac{\sqrt{3}}{2}\tilde{s}_1-\frac{1}{2}S_2,\quad s_3=-\frac{\sqrt{3}}{2}\tilde{s}_1-\frac{1}{2}s_2 \tag{5}$$

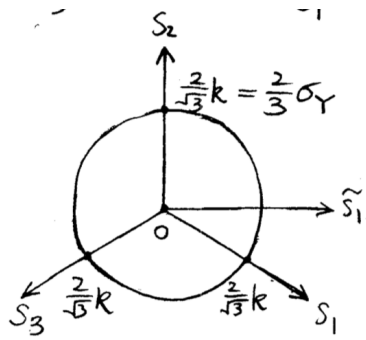
Substitute into Eq. (2):

$$k^2 = J_2 = \frac{1}{2}(s_1^2 + s_2^2 + s_3^2) = \frac{3}{4}(\tilde{s}_1^2 + s_2^2) \tag{6}$$

This indeed simplifies to the equation of a circle:

$$\tilde{s}_1^2 + s_2^2 = \left(\frac{2}{\sqrt{3}}\,k\right)^2 = \left(\frac{2}{3}\sigma_Y\right)^2 \tag{7}$$

The Von Mises yield condition is a circle of radius $\frac{2}{\sqrt{3}}k = \frac{2}{3}\sigma_Y$ in the π-plane of s_1, s_2, s_3.



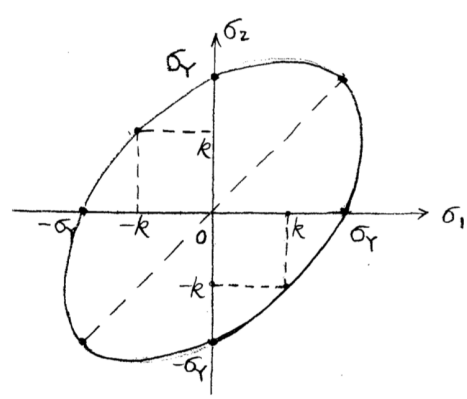
Von Mises criterion in plane stress ($\sigma_3 = 0$)

Under plane stress condition ($\sigma_3 = 0$),

$$\begin{aligned} J_2 &= \frac{1}{6}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right] \\ &= \frac{1}{6}(\sigma_1^2 - 2\sigma_1\sigma_2 + \sigma_2^2 + \sigma_2^2 + \sigma_1^2) \\ &= \frac{1}{3}(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2) = k^2 \end{aligned} \tag{8}$$

So the intersection between the Von Mises yield surface and the $\sigma_3 = 0$ plane is a curve described by the following equation, which describes an ellipse.

$$\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2 = (\sqrt{3}k)^2 = \sigma_Y^2 \tag{9}$$



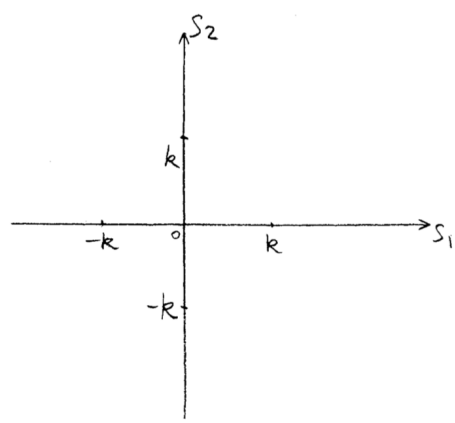
Special points:

- Pure tension: $\sigma_1 = \sigma_{\mathrm{Y}}, \quad \sigma_2 = 0$
- Pure shear: $\sigma_1 = k, \quad \sigma_2 = -k$
- Biaxial tension: $\sigma_1 = \sigma_{\mathrm{Y}}, \quad \sigma_2 = \sigma_{\mathrm{Y}}$

Sometimes, the Von Mises yield condition is sketched in the s_1-s_2 plane. By definition $s_1 + s_2 + s_3 = 0$, so that $s_3 = -(s_1 + s_2)$. Hence

$$\begin{aligned} J_2 &= \frac{1}{2}\left(s_1^2 + s_2^2 + s_3^2\right) \\ &= \frac{1}{2}\left(s_1^2 + s_2^2 + (s_1 + s_2)^2\right) \\ &= s_1^2 + s_2^2 + s_1 s_2 = k^2 \end{aligned} \tag{10}$$

Exercise: Sketch the Von Mises yield surface on the s_1-s_2 plane.



3 Tresca yield condition

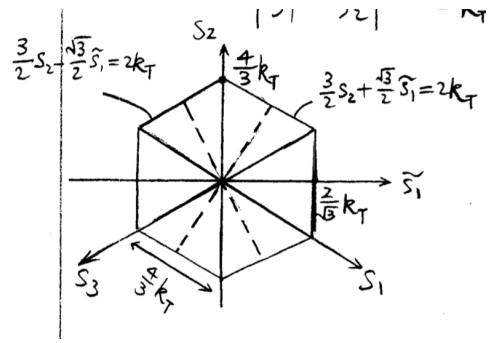
The Tresca yield condition is defined by the maximum shear stress:

$$|s_1 - s_2| = 2\,k_{\mathrm{T}}, \quad \text{or} \quad |s_2 - s_3| = 2\,k_{\mathrm{T}}, \quad \text{or} \quad |s_3 - s_1| = 2\,k_{\mathrm{T}} \tag{11}$$

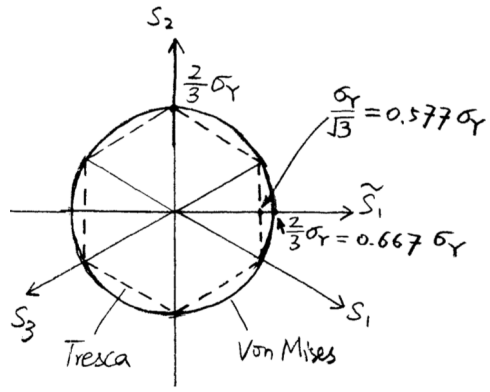
In terms of $\tilde{s}_1$ and s_2:

$$\left|\frac{\sqrt{3}}{2}\tilde{s}_1 - \frac{3}{2}s_2\right| = 2\,k_{\mathrm{T}}, \quad \text{or} \quad \left|\frac{\sqrt{3}}{2}\tilde{s}_1 + \frac{3}{2}s_2\right| = 2\,k_{\mathrm{T}}, \quad \text{or} \quad \left|\sqrt{3}\,\tilde{s}_1\right| = 2\,k_{\mathrm{T}} \tag{12}$$

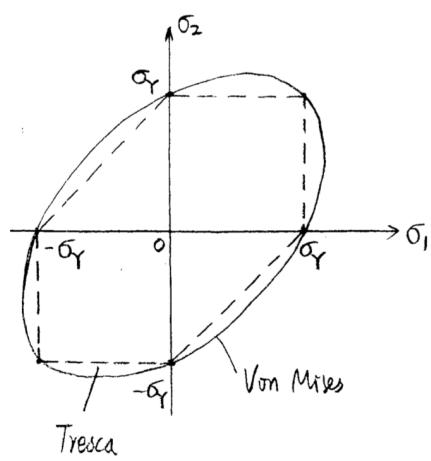
The Tresca yield condition forms a hexagon of side length $\frac{4}{3}\,k_{\mathrm{T}} = \frac{2}{3}\,\sigma_{\mathrm{Y}}$ in the deviatoric plane (s_1, s_2, s_3).



On this plane, the Tresca yield condition is a hexagon inscribed in the Von Mises circle.



For plane stress ($\sigma_3 = 0$), the Tresca condition $|\sigma_1 - \sigma_2| = \sigma_\mathrm{Y}$, or $|\sigma_1| = \sigma_\mathrm{Y}$, or $|\sigma_2| = \sigma_\mathrm{Y}$ forms a polygon (dashed line) inscribed in the Von Mises ellipse (solid line) as shown below.



4 Flow rule for perfectly plastic material

In this section, we provide a graphical representation of how the stress and strain tensors evolve during the flow of a perfectly plastic material (following Hill 1951).

Elastic response:

The deviatoric part of the elastic strain is proportional to the deviatoric part of the stress. Hence their increments are also simply proportional to each other.

$$e_{ij}^{\mathrm{el}} = \frac{1}{2\mu} s_{ij} \implies de_{ij}^{\mathrm{el}} = \frac{1}{2\mu} ds_{ij} \tag{13}$$

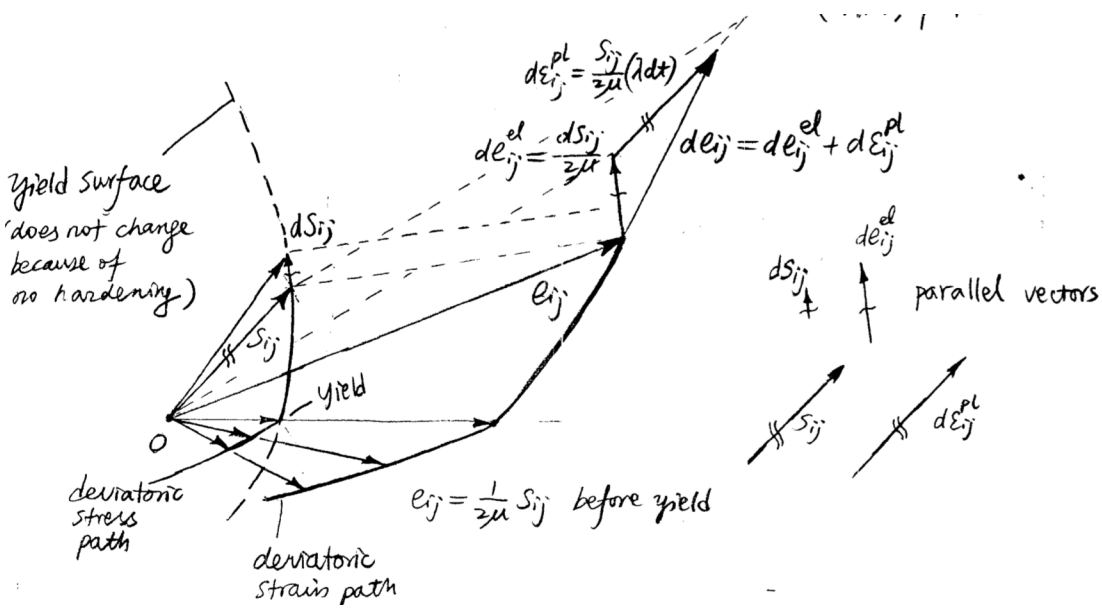
Plastic response: The plastic strain rate is proportional to the deviatoric part of the stress. Hence the increment of the plastic strain tensor is proportional to the deviatoric stress.

$$\dot{\varepsilon}_{ij}^{\mathrm{pl}} = \frac{1}{2\mu} \tilde{\lambda}\, s_{ij} \implies d\varepsilon_{ij}^{\mathrm{pl}} = \frac{s_{ij}}{2\mu} (\tilde{\lambda}\, dt) \tag{14}$$

Therefore, the total deviatoric strain increment is:

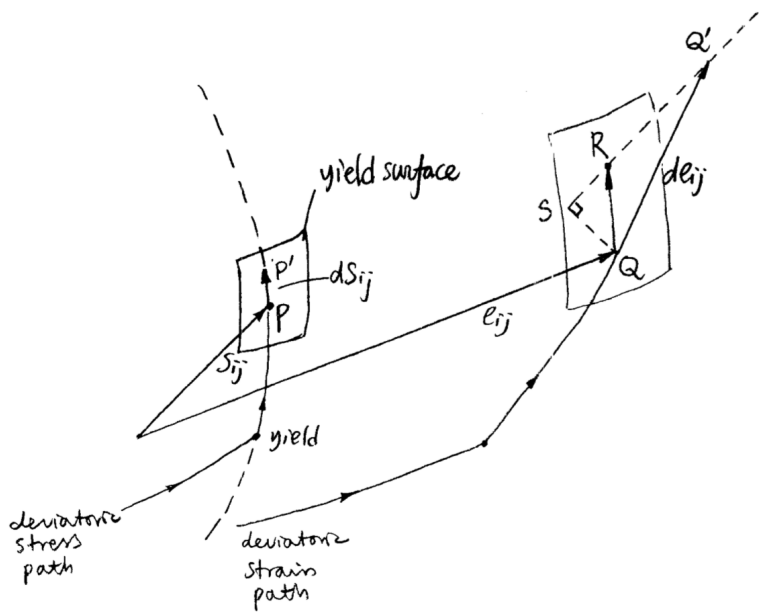
$$de_{ij} = de_{ij}^{\mathrm{el}} + d\varepsilon_{ij}^{\mathrm{pl}} = \frac{ds_{ij}}{2\mu} + \frac{s_{ij}}{2\mu} (\tilde{\lambda}\, dt) \tag{15}$$

The above relationships can be graphically represented as follows (similar to Hill, Fig.7, p.42), using arrows to illustrate tensors. Parallel arrows indicate tensors that are proportional to each other (i.e. along the same direction).



Geometric interpretation:

- Before s_{ij} reaches the yield surface, e_{ij} is parallel to s_{ij}, and de_{ij} is parallel to ds_{ij}.
- After the yield condition is reached, e_{ij} is no longer parallel to s_{ij}, and de_{ij} is no longer parallel to ds_{ij}.
- Instead, $de_{ij} = de_{ij}^{\mathrm{el}} + de_{ij}^{\mathrm{pl}}$. de_{ij}^{el} is parallel to ds_{ij} (generalized Hooke's law), while de_{ij}^{pl} is parallel to s_{ij} (normality rule).



Graphical construction:
In the plastic regime, given s_{ij}, e_{ij}, and the total deviatoric strain increment de_{ij}, we want to find ds_{ij} (change of deviatoric stress).

1. Draw a line SQ' parallel to s_{ij}.
2. Construct the patch of the yield surface at point P.
3. Draw a parallel surface patch at point Q.
4. Find the intersevtion point R between line SQ' and the surface patch.
5. As a result, vector RQ' $= d\varepsilon_{ij}^{\mathrm{pl}}$ and vector QR $= de_{ij}^{\mathrm{el}}$, and vector PP' $= ds_{ij} = 2\mu \cdot$ QR.